AVIRAL MARWAL

Assistant Professor of Economics | Department of Humanities and Social Sciences
Indian Institute of Technology Roorkee, Uttarakhand 247667, India
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RESEARCH INTERESTS

Urban economics and urban policy; spatial analysis using GIS and R; geo-computational simulation with agent-based and cellular automata models; residential location choice and travel demand modelling; accessibility, land use and urban form.

EDUCATION

PhD, Land Economy 2020 – 2023
University of Cambridge, United Kingdom
Thesis: Examining the Dynamics of Urban Form, Flow, and Accessibility Using Geo-Computational Methods — A Case Study of Delhi. Supervisor: Prof. Elisabete A. Silva.

MSc, Development Policy, Planning and Practice 2014 – 2016
Tata Institute of Social Sciences, Mumbai (GPA 7.2/10)
Thesis: Understanding the service delivery mechanism of urban local bodies in India — solid waste management in Ajmer.

BTech, Engineering Physics 2010 – 2014
Indian Institute of Technology Delhi (GPA 6.6/10)

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Assistant Professor (Economics) Nov 2024 – Present
Department of Humanities and Social Sciences, IIT Roorkee
Teaching, doctoral supervision and sponsored research in urban and institutional economics.

Research Scientist Apr 2024 – Oct 2024
CiSTUP, Indian Institute of Science (IISc) Bangalore
Population synthesis and travel demand forecasting using an agent-based approach.

Senior Research Associate Oct 2023 – Mar 2024
Sankala Foundation, New Delhi
Policy lessons from the Jal Jeevan Mission for India and the Global South.

Research Assistant, Department of Economics Aug 2019 – Sep 2020
Indian Institute of Management Bangalore
With Prof. Gopal Naik: enhancing women's livelihood opportunities through innovative financing mechanisms in Tamil Nadu.

SPONSORED RESEARCH PROJECTS

Assessment of Financial Resources of Local Self-Governing Institutions (PRIs and ULBs) in Uttarakhand May 2025 – Present
Co-Principal Investigator · SETU Aayog, Government of Uttarakhand · INR 27 lakh
Assessing the financial position of PRIs and ULBs to determine the divisible pool for State financial devolution.

Neighbourhood Accessibility and Gentrification Sep 2025 – Present
Principal Investigator · SRIC, IIT Roorkee · INR 13 lakh

PUBLICATIONS

Peer-reviewed journal articles

1. Marwal, A., & Silva, E. A. (2024). Simulating built-up expansion in west Delhi using a neural network coupled agent based prioritised growth model. *Letters in Spatial and Resource Sciences*, 17, 31. Springer.
2. Marwal, A., & Silva, E. A. (2024). Analysing inequity in accessibility to services with neighbourhood location and socio-economic characteristics in Delhi. *Geographical Analysis*. Wiley (Q1, IF 3.6).
3. Marwal, A., & Silva, E. A. (2023). Exploring residential built-up form typologies in Delhi: a grid-based k-means clustering approach mapping sustainable urbanization. *npj Urban Sustainability*. Springer Nature (Q1, IF 9.1).
4. Marwal, A., & Silva, E. A. (2023). City affordability and residential location choice: a demonstration using agent-based model. *Habitat International*, 136, 102816. Elsevier (Q1, IF 6.8).
5. Marwal, A., & Silva, E. A. (2022). Literature review of accessibility measures and models used in land use and transportation planning in the last five years. *Journal of Geographical Sciences*, 32(3), 560–584. Springer (Q1, IF 5.7).

Conference proceedings, case studies and working papers

1. Pinjari, A. R., Rambha, T., Koushik, A., Marwal, A., et al. (2025). Aggregate-M3DM: a city-scale multi-modal mobility demand modelling tool for Indian cities. 8th Conference of the Transportation Research Group of India (CTRG), IIT Guwahati.
2. Marwal, A., Naik, G., & Dalhia, M. (2020). The Saboo Foundation model of philanthropy: how a self-made man created thousands more. Case study IMB 844, Harvard Business Publishing.
3. Marwal, A., & Silva, E. A. (2024). Commuting behaviour, built environment and travel attitude: insights from a field survey in Delhi. Working paper, SSRN.

TEACHING

Indian Institute of Technology Roorkee 2025 – Present
Assistant Professor, Department of Humanities and Social Sciences
Institutional Economics (HS 508, Spring); Urban Economics (HS 535, Autumn); Spatial Data Analysis (HS 659, Spring); Welfare Economics (HS 601, Autumn).

University of Cambridge 2021 – 2023
Supervisor and Tutor, Department of Land Economy
Business Economics (UG and PG): demand estimation, consumer choice, market structure and pricing, game theory, market failure. Land and Urban Economics (PG): agent-based modelling and GIS. Quantitative Research Methods (UG): statistics, basic econometrics, linear programming.

King Saud University, Riyadh Jul – Aug 2022
Guest Faculty, summer school
Introduction to Economics; jointly organised by the Government of Saudi Arabia and Oxmedica Pvt. Ltd.

Lumiere Education Jun 2023 – Sep 2024
Research Supervisor (part-time)
Supervised seven students through research paper writing and publication.

East-West Education Group, United Kingdom Oct 2020 – Jul 2022
Course Facilitator (part-time)
Delivered academic essay-writing modules to undergraduate students.

Doctoral Supervision

Mr. Nikhil Mishra Since Jan 2025
Firm Agglomeration and Urbanisation

Ms. Ankita Dey Since Jul 2025
Urban Land Use Policy

Research Collaborations

— Population synthesis for agent-based travel demand models — CiSTUP, Indian Institute of Science Bangalore.

— Mapping opportunities for healthy and sustainable urban transitions in post-industrial cities — Urban Studies Foundation, United Kingdom.

Conferences, Seminars and Events

RSA Annual Conference: Regions as Arenas in a Changing World, Sweden 14 – 18 Jun 2026
Funded by IIT Roorkee

Conference on Cities and Urban Planning in Ancient India 23 – 25 Dec 2025
Organised by IIT Indore

Organised seminar on Mapping New Directions in Urbanisation: Challenges and Emerging Futures 9 Dec 2025
Funded by the Urban Studies Foundation

Conference of the International Association of Mathematical Geosciences (IAMG), Norway 1 – 4 Aug 2023
Funded by the University of Cambridge

Papers presented and invited talks

1. Utilizing ABM and neural network models to simulate built-up expansion in Delhi. International Association of Mathematical Geosciences (IAMG), NTNU Trondheim, Norway (August 2023).
2. A study of the commuting pattern of households in Delhi. Regional Science Association International (RSAI), University of Newcastle, United Kingdom (July 2023).
3. Invited open lecture on gentrification and housing affordability, O. P. Jindal Global University (April 2024).
4. An analytical framework to study urbanisation using spatial data, CiSTUP, IISc Bangalore (October 2023).
5. Building geospatial data from satellite imagery to study urban form, Urban Big Data Centre, University of Glasgow (September 2023).

Field Research

PhD fieldwork, Delhi (September 2021 – February 2022). Designed and led a door-to-door household survey of 1,700 respondents across income categories with a team of 15 field assistants, capturing travel behaviour and residential location choice.

Honours, Memberships and Administrative Roles

— Swami Vivekanand Fellowship for Academic Excellence, Government of Rajasthan (2020).

— GRE 166/170 (Quantitative); TOEFL 109/120; UGC NET 99.7 percentile.

— Member, Regional Science Association International, United Kingdom (since 2021); Member, International Association of Mathematical Geosciences, Norway (2023–24).

— Convenor, Departmental Faculty Search Committee, HSS, IIT Roorkee (since January 2026); Faculty Advisor, Economics Circle, IIT Roorkee (since June 2026).

— Lead, Research Lab on City Experiments and Learning, IIT Roorkee.

Technical Skills

Programming: R, Python. Software: ArcGIS, QGIS, NetLogo, Stata, SPSS. Methods: agent-based and cellular automata modelling, machine learning for spatial prediction, survey design, econometric analysis.

References

Prof. Elisabete A. Silva
PhD Supervisor · Department of Land Economy, University of Cambridge · es424@cam.ac.uk

Prof. Shaun Larcom
Department of Land Economy, University of Cambridge · stl25@cam.ac.uk

Prof. Eric Heikkila
Sol Price School of Public Policy, University of Southern California · heikkila@usc.edu